Rational curiosity predicts the dynamics of habituation and dishabituation in infants and adults

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2023-11-13

1 Abstract

How do we decide what to look at and when to stop looking? Even very young infants engage in active visual selection, looking less and less as stimuli are repeated (habituation) and regaining interest when novel stimuli are subsequently introduced (dishabituation). The mechanisms underlying these looking time changes remain uncertain, however, due to limits on both the scope of existing formal models and the empirical precision of measurements of infant behavior. We develop the Rational Action, Noisy Choice for Habituation (RANCH) model, which operates over raw images and makes quantitative predictions of participants' looking behaviors. In a series of pre-registered experiments, we measure infants' and adults' looking time across a range of stimulus exposure durations for both repeated and novel stimuli and show that they are predicted by RANCH. Our model also makes out-of-sample predictions about the magnitude of adult and infant dishabituation to a range of novel stimuli in an independent experiment. By framing looking behaviors as rational decision-making, this work identifies how the dynamics of learning and exploration guide our visual attention from infancy through adulthood.

2 Introduction

From birth, humans engage in active learning. Even before they gain independent mobility, infants make choices about what to look at and when to stop looking [20, 40]. Developmental psychologists have both studied this process of visual decision-making as a phenomenon itself and also relied on it as a key methodological tool. Two key phenomena have been especially important: habituation and dishabituation. Habituation occurs when infants show reduced interest upon repeated exposure to the same stimulus, whereas dishabituation entails renewed interest following the subsequent introduction of a novel stimulus. Dishabituation, also known as novelty preference, is often leveraged to infer infants' perceptual and cognitive abilities [2, 3, 16]: in order for an infant to dishabituate, they must distinguish between the original stimulus and the novel one. Despite the robust documentation of habituation and dishabituation in the literature, the mechanisms underlying these changes in gaze duration remain poorly understood. In this paper, we bridge this gap by introducing a rational model of habituation and dishabituation. We show that this rational model makes quantitative predictions of looking times toward different stimuli.

An important early conceptual model of infant looking time posited that habituation and dishabituation are influenced by the amount of information to be encoded by the infant from the stimulus [25]. Initially, infants are assumed to look longer at stimuli containing substantial unprocessed information, showing a familiarity preference (Figure 1A). As exposure accumulates, less information remains unprocessed, leading to shorter looking time at the original stimulus and by comparison, relatively longer looking time to a new stimulus (i.e. novelty preference). While this model has had significant influence, the absence of specifications regarding the encoding process or a way of measuring the amount of information to be encoded in a stimulus have both permitted post-hoc explanations of variable looking time measurements. For instance,

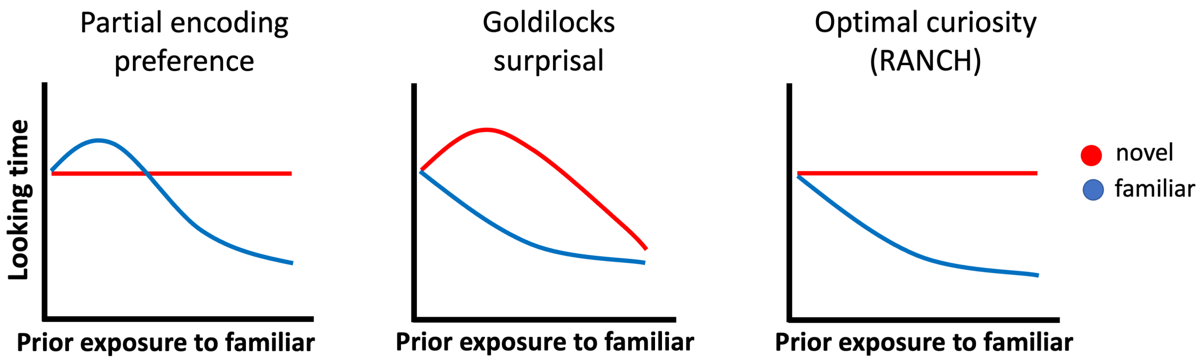


Figure 1: Conceptual models of looking to novel vs. familiar items according to three models of infant attention: A) The partial encoding model proposed by Hunter and Ames (1988) which suggests a shift from familiarity to novelty preferences, B) The Goldilocks model proposed by Kidd et al. (2012) which suggests that infants prefer to attend to intermediately surprising events, and C) the 'optimal curiosity' model introduced by Cao et al. (2023) which suggests that infants are maximizing expected information gain from noisy perceptual samples.

when infants are looking longer at stimulus A than stimulus B, researchers could interpret the looking time difference as either a novelty preference or a familiarity preference. This interpretive ambiguity has spurred repeated concerns regarding whether looking time measurements should form the basis for key assertions in developmental psychology [7, 21, 36]. In other words, developmental psychology needs formal models that make quantitative predictions, beyond the qualitative foundations of Hunter and Ames [25].

Recent computational models aim to formalize the mechanisms underlying looking time changes. One family of models characterize infants' looking behaviors through information-theoretic metrics derived from ideal observer models: the models acquire probability distributions from event sequences and derive information-theoretic measures from the ideal learner's belief before and after each event [28, 37, 38]. For instance, Kidd, Piantadosi, and Aslin [28] found that an ideal learner's surprisal at an event was related to infants' looking behavior via a U-shaped curve: infants were least likely to look away when the surprisal of the event was neither too high nor too low (1B, model predictions derived in 8).

While these models provide quantitative fits to variation in looking time, they are not models of how an agent makes the decisions of whether to keep looking at the same stimulus or to look at something else. Rather than producing the target behavior directly, these models describe correlations between model-derived, information-theoretic measures and infants' overall looking towards whole sequences of events. Relatedly, these models do not explain why infants would ever look at one single stimulus for a long time, rather than encode it instantaneously – there is no account of perceptual noise to be overcome by taking multiple perceptual samples that would justify why looking extends over time [c.f. 8, 27]. Owing to this constraint, these models can only be used to predict infant behavior in a specific paradigm: looking to a continuous stream of discrete stimuli. This paradigm deviates significantly from the more common paradigm in which infants gradually encoded individual objects or events in a self-paced manner. Thus, existing models are challenging to generalize to more common looking time paradigms.

To address these issues, we developed the Rational Action, Noisy Choice for Habituation (RANCH) model [9]. RANCH describes an agent's looking behavior as rational exploration based on a sequence of noisy perceptual samples. The model construes the looking time paradigm as a series of binary decisions: to keep sampling from the current stimulus, or to look at anything else. The model makes sampling decisions based on the Expected Information Gain (EIG) of the upcoming perceptual sample, choosing to keep looking or look away based on which one would in expectation yield the most information. The RANCH model is therefore a rational analysis of looking behavior [32, 34, 1]: looking is analyzed as optimally adapted to the environment, given specific constraints on processing (e.g. perceptual noise).

Previously, we validated this rational analysis using a large dataset of adults engaging in a self-paced looking paradigm. We asked adults to look at sequences of stimuli consisting of a repeating familiar background stimulus (resulting in habituation), and a single, novel deviant stimulus after varying familiarization durations (resulting in dishabituation). Effects of prior exposure and novelty on adults' looking time were well captured by RANCH [9]. Figure 1C shows schematic predictions of RANCH for the experimental setting described above: variable prior exposure to a familiar stimulus, followed by a novel or a familiar stimulus. Rational exploration of a simple perceptual concept results in a monotonic decrease of attention to familiar items as a function of prior exposure.

In our work, we address three challenges to adjudicate between the candidate models of infant looking in Figure 1. A first critical challenge for differentiating formal theories of looking time in infants is that looking time measurements are noisy, and many infant experiments provide only limited precision in their estimates of key quantities [10, 11]. In the current work, we develop a novel, within-subject experimental design that allows us to sensitively measure infants' looking time in experiments. In our first experiment, we use this paradigm to measure the dynamics of habituation and dishabituation across different amounts of exposure to a stimulus. Beyond validating RANCH's predictions in infants, this paradigm enables us to conduct interpretable developmental comparisons of parameter fits between infants and adults – since we can conduct matched experiments in adults – thereby characterizing what aspects of the processes underlying looking time may change over development.

A second critical challenge for differentiating theories has been their stimulus-dependence. Infancy researchers can rarely make quantitative predictions about differences between stimuli; even previous computational models have relied on using arbitrary representations that are not directly generated from the stimulus materials [e.g. object location, 28, hand-specified binary feature vector, 9]. To address this issue, we incorporated recent progress in convolutional neural networks (CNNs) to allow RANCH to learn from raw images. The representations of many trained CNNs show striking similarities to how primate brains respond to stimuli, and investigating the representations of CNNs has therefore offered insights into how the visual system encodes objects [13, 23, 46]. The activations of these brain-inspired neural networks form embedding spaces, each of which can be seen as a quantitative hypothesis about how humans embed objects in a lower-dimensional perceptual space [41]. In RANCH, we leveraged these principled stimulus representations to enable RANCH to learn from raw images. In other words, RANCH can 'see' an experiment in a way similar to the one infants saw, and make predictions for previously unseen experiments using novel stimuli.

A final challenge is the generalizability of models to new contexts. Because RANCH can be applied across stimulus sets without additional assumptions, we are able to test its predictions in a new dataset. To conduct these tests, we first fit RANCH's parameters to a training data set from the initial habituation-dishabituation experiment. Then, we use the best-fitting parameters to generate predictions for a new experiment designed to measure differences in dishabituation magnitude. In this experiment, we systematically varied the similarity between habituation and dishabituation stimuli such that dishabituation stimuli differed in their pose angle, their number, their identity, or their animacy. This experiment tests a prediction from Hunter and Ames [25] model of habituation and dishabituation: that observers' dishabituation magnitude should be related to the similarity between the habituated stimulus and the novel stimulus. The more dissimilar two stimuli are, the more one should dishabituate to the novel stimulus.

Addressing these challenges allowed us to compare the different accounts of habituation and dishabituation in 1 empirically. To preview our results, we find that RANCH provided the best description of behavior in our first experiment, in which we systematically vary prior exposure to a stimulus before measuring looking time to either the same, familiar stimulus or a novel stimulus. Fitting RANCH to infant and adult data separately and comparing the best-fitting parameters revealed that infants' learning was tuned for noisier processing than adults. Then, using these best fitting parameters, we generate predictions for a novel task in which we vary the similarity between the familiar and novel stimulus, and find that RANCH generalizes to a novel task context by predicting the dishabituation magnitude as a function of stimulus dissimilarity. Together, we provide evidence that habituation and dishabituation are well described by a rational analysis of looking time.

3 Rational Action, Nosy Choice for Habituation model

The Rational Action, Noisy Choice for Habituation model (RANCH) is a Bayesian perception and action model in which a learning model is used to derive optimal perceptual sampling decisions [9]. In RANCH, the learning problem in habituation is conceptualized as learning a visual concept – a single representation underlying a group of observed stimuli. The model learns by observing a series of noisy perceptual samples from each stimulus. During learning, the model makes decisions about how many samples to receive from each stimulus before moving to the next stimulus. We treat the number of samples for each stimulus as being linearly related to looking time duration.

RANCH provides a probabilistic framework for making perceptual decisions. In this section, we will describe the three modular components of RANCH: the perceptual representation, learning model, and decision model.

3.1 Perceptual representations

We use the deep convolutional neural network (CNN) ResNet-50 to encode the stimuli used in behavioral experiments. ResNet-50 is a standard architecture that is widely used in computer vision as well as in comparisons to human behavior [22, 30]. A total of 50 layers are implemented in this architecture: it starts with a convolution layer, includes several groups of residual blocks that help preserve information through the network, and ends with a global average pooling layer and a fully connected layer for classification. We used a version of ResNet-50 that was pretrained on ImageNet [12]. Training on ImageNet optimizes the Resnet-50's activations for image classification, and we can use these activations as an perceptual "embedding space". When we pass new stimuli through a pre-trained network, we extract these activations to obtain principled embeddings of stimuli (Figure 2A). We used the first three principal components of the embedding space to limit the dimensionality of the downstream learning problem.

[1]

3.2 Learning model

RANCH models looking behaviors in habituation experiments as Bayesian concept learning [19, 44]. The model infers the concept that is most likely to have generated a series of noisy perceptual samples taken from a sequence of stimuli. The concept to be learned is a multivariate normal distribution parameterized by μ, σ, which represents beliefs about the location and variance of the presented concept in the embedding space. The concept μ, σ generates exemplars y, which is equivalent to the stimulus seen by the human participants. But RANCH does not directly learn from the stimulus. Instead, it learns from the series of noisy perceptual samples $\bar{z}$ generated by the exemplar. The noisy perceptual sample is corrupted by zero-mean gaussian noise, represented by ϵ. A plate diagram is shown in Figure 2B. At each time step, the model takes one perceptual sample to update the prior following Bayes Rule:

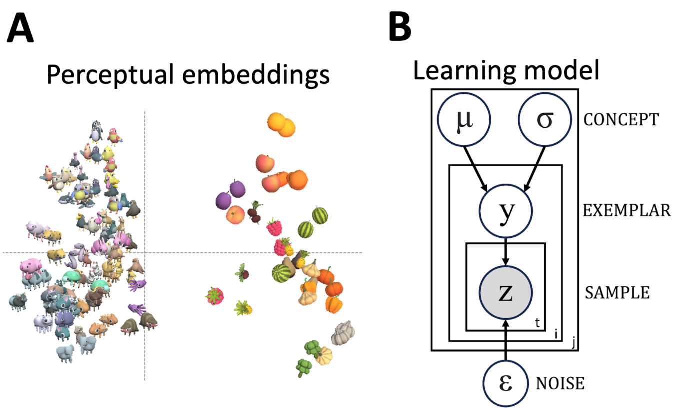
$$p(\mu, \sigma|\bar{z}) = \frac{p(\bar{z}|\mu, \sigma) * p(\mu, \sigma)}{p(\bar{z})}$$

3.3 Decision model and linking hypothesis

The Bayesian learning model does not address how the model decides to keep sampling the same stimulus or to move on to the next stimulus.To make this decision, RANCH uses the the expected information gain (EIG) of

[1]ResNet-50 is by no means the only possible method for deriving perceptual representations; instead it is a simple, standard choice. We also explored alternative models, such as a CNN trained on egocentric videos from the child's perspective and CNN whose representations were aligned to human categorization behavior [35, 31]. However, we did not find substantial differences between the embedding spaces provided by ResNet-50 and other (often more complicated) models (Figure 12).

the next perceptual sample. EIG is a metric used under the rational analysis of information-seeking behavior [1, 34], and is computed as the expected value of the KL divergence between the hypothetical posterior at the next timepoint $t + 1$ and the current posterior. Formally: $EIG = \mathbb{E}_{\approx+\not\models}(KL(P(\mu, \sigma|z_{t+1}), P(\mu, \sigma|z_t)))$. RANCH then uses a softmax choice (with temperature = 1) between the information theoretic measures of the next sample and a constant, which is assumed to represent the amount of information gain provided by the environment when the learner looks away from the stimulus. An algorithm table describing the overall sampling procedure is shown in Figure 2C.



C

RANCH sampling algorithm

RANCH algorithm

1: **for** each exemplar y **do**
2:   *sample* ← T
3:   **while** *sample* is T, take another sample z
4:     update posterior $P(\mu, \sigma|z)$
5:     compute EIG of next sample z_{t+1}
6:     flip *coin* with $p(lookaway) = \frac{EIG(env)}{EIG(z_{t+1})+EIG(env)}$
7:     **if** $coin = T$
8:       *sample* ← F
9:     **end if**
10:   **end while**
11: **end for**

Figure 2: Three components of RANCH: A) Perceptual representation: stimulus embeddings from ResNet-50 final layer activations, B) Learning model: plate diagram for Gaussian concept learning from noisy observations, C) Decision model: RANCH repeatedly samples until environmental EIG outweighs EIG from another sample.

3.4 Implementation

RANCH has free parameters including the learner's priors over the concept to be learned (μ, σ), the learner's priors over their perceptual noise (σ_ϵ) and the constant representing EIG from the environment (EIG_{env}). We conducted an iterative grid search over these parameters.

While gaussian concept learning models can be computed analytically, hierarchical models that include perceptual noise must be approximated via sampling methods. We used grid approximation, creating discrete grids over μ, $sigma$, y (exemplars) and z (noisy samples) to perform inference. We further used a grid over possible next observations when computing EIG.

To remove noise resulting from this grid approximation, we averaged each stimulus and parameter setting over 100 runs. To avoid bias, these 100 simulations used 20 different "jitters' ' - such that the exact location of the points in the discrete grids was randomly chosen. We used the number of samples the model decided to take from each stimulus as a proxy for looking time. We then averaged this metric across the 100 runs.

This process required significant computational resources. The grid approximation, combined with repeated simulations, demanded high-performance cluster GPUs to perform vectorized computation using a PyTorch implementation of our model. Running a single parameter setting through an experiment took approximately 10 hours on a GPU. By distributing different parameter settings across GPUs, a full parameter search took about 3 days of compute.

4 RANCH predicts habituation and dishabituation in relation to exposure duration

The models in Figure 1 make contrasting predictions with regard to the effect of increasing prior exposure on looking times. To directly test these predictions, we collected looking time data from both infants and adults in an experiment where we varied cumulative exposure to a familiar stimulus, and measured subsequent looking time to a novel or familiar stimulus. We then let RANCH "see'' the same stimulus sequences used in our behavioral experiments and compared our model's behavior to that of humans.

4.1 Behavioral experiments

4.1.1 Infant experiment

We tested infants in a novel experimental paradigm in which exposure duration to the familiar stimulus was manipulated in a within-subject design 3 (Experiment 1, Infants). Each infant saw six blocks. In each block there was an familiarization phase and a test trial. During the familiarization phase, one animal (designated the familiar stimulus) was presented 1 to 9 times. Next, infants' looking time was measured on a test trial. The test trial showed either the familiar stimulus, or a novel animal. We ran three versions of this experiment, each with three different numbers of familiarization events. We tested a combined sample of 103 7-10 month old infants, with 31 infants in Exp 1A (0, 4 or 8 familiarization events), 35 infants in Exp 1B (1, 3 or 9 familiarization events) and 37 infants in Exp 1C (2, 4 or 6 familiarization events; total n=103, M_{age}=9.38 months, 47 female).

4.1.2 Adult experiment

We ran a self-paced looking time experiment on Prolific, collecting data from 470 adult participants 3 (Experiment 1, Adults). Stimuli used in the adult experiment were drawn from the same set used in the infant experiment. During the experiment, participants were asked to watch sequences of stimuli at their own pace, and they could press a key to proceed to the next stimulus whenever they wanted to. The time between the onset of the stimulus and the key press was used as a proxy for looking time. Each sequence began with a familiar stimulus repeating 1 to 10 times, followed by either the same familiar or a novel stimulus.

4.1.3 Modeling Procedure

RANCH received the CNN-derived stimulus embeddings as input, presented in sequences mirroring the structure of in our infant and adult experiments (3). To simulate a block of the infant experiment, the model was first presented with the "familiar" stimulus. For each stimulus, the model took 5 samples each and then moved on to the next stimulus. This phase was designed to be to the familiarization phase infants went through, since during the familiarization phase infants had no control over the progression of the stimulus sequence. After the familiarization phase, the model was presented with the test trial, which was either familiar or novel. We used the number of samples the model took on this test trial as a proxy for infant looking time. To simulate the self-paced adult experiment, we computed the number of samples taken by RANCH on every trial in each stimulus sequence.

4.2 Behavioral and model results

The paradigm successfully captured habituation in both infants and adults. In infants, as the number of trials in exposure increased, the looking time at the familiar stimulus in the test trials decreased (β = -4.55; SE = 0.92; t = -4.92; p < 0.001). Infants clearly dishabituated on trials longer exposures (8 exposures: β

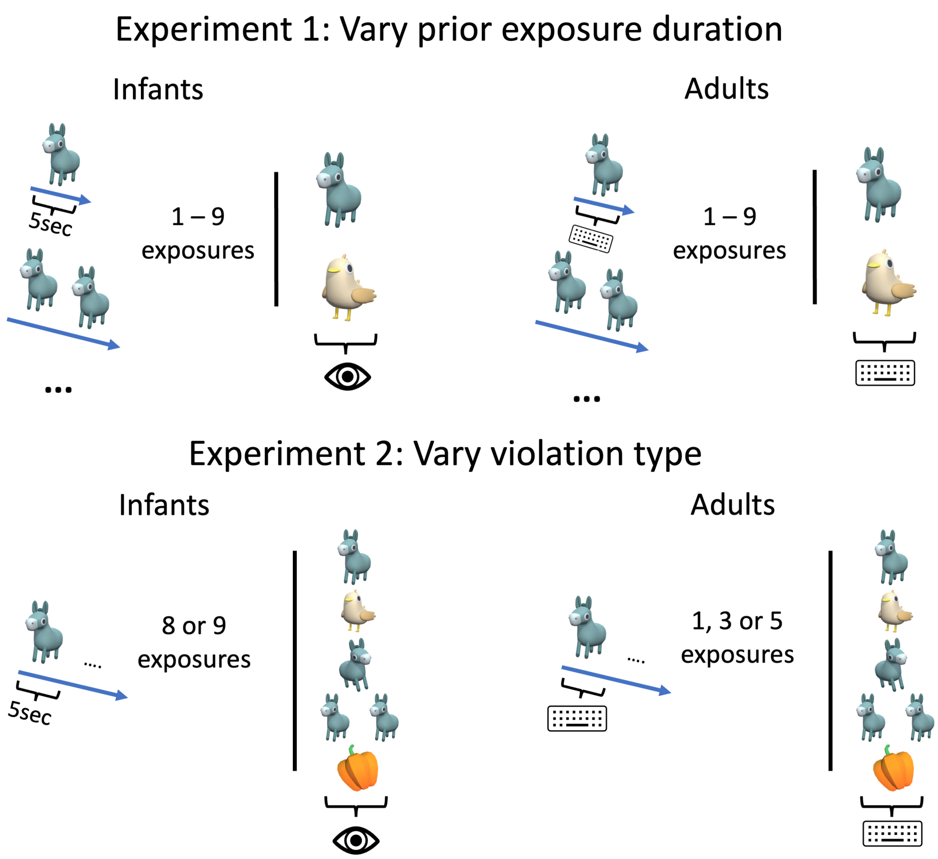


Figure 3: Experimental design

Table 1: Cross-validated RMSE across linking hypotheses and lesioned models. Mean RMSE is averaged across all parameter values, best RMSE is the single best-performing parameter set

Linking Hypothesis	Infants: Mean RMSE	Infants: Best RMSE	Adults: Mean RMSE	Adults: Best RMSE
EIG	3.939	3.696	0.222	0.166
KL	3.912	3.658	0.224	0.163
Surprisal	4.396	3.601	0.303	0.213
No learning	4.463	4.135	0.449	0.377
No noise	4.224	3.875	0.339	0.292

= 0.82; SE = 0.22; t = 3.81; $p < 0.001$, 9 exposures: $\beta = 0.51$; SE = 0.12; t = 4.21; $p < 0.001$). The adult experimental paradigm also successfully captured habituation and dishabituation (Trial number: β = -0.01; Trial Type: β = 0.22; Both $SE < 0.001$; Both $p < 0.001$). We found no evidence for familiarity preferences or non-linearities in the shape of the dishabituation curve in infants or adults (cf. 1A&B).

RANCH predicted habituation and dishabituation in both infants and adults. To compare model samples to looking times, we fit a linear model adjusting the number of samples by an intercept and a scaling factor. We then computed the root mean squared error (RMSE) between the scaled model results and the looking time data. To avoid overfitting on these scaling parameters and the free parameters that we searched over, we used K-fold cross-validation, computing the average RMSE across held-out samples. This procedure was conducted separately for both infant and adult data.

4.3 Comparison with alternative linking hypotheses

EIG is the optimal method for making sampling decisions [33]. We compared this method to decision-making metrics proposed in prior literature including Kullback-Leibler divergence (KL-divergence) and surprisal. KL-divergence measures the information gained when one's beliefs are updated, effectively quantifying "learning progress", while surprisal captures how inconsistent incoming information is with prior expectations. Both have been shown to be associated with infants' looking behaviors [28, 37]. We then repeated the cross-validation process using these linking hypotheses and compared the model's performance based on the parameter that provides the best average fits to the test sets according to RMSE. Out of the three linking hypotheses, we found that EIG and KL-divergence were comparable to each other but surprisal-based RANCH performed worse in both populations (Table 1).

4.4 Comparison with lesioned model

We tested the importance of two of RANCH's key assumptions by creating lesioned models. One assumes each perceptual sample is noiseless ("No noise"), and the other assumes the sampling decision was random instead of based on the learning model ("No learning"). Both lesioned models provided poor fits to the behavioral datasets (Table 1). Both the noisy perception assumption and the learning assumption appear to be critical to RANCH's performance.

4.5 Parameter Interpretation / Developmental Comparison

One advantage of RANCH is the interpretability of its parameters. Beyond finding that RANCH's fit to the data is robust across parameters, we can also ask which specific parameters maximize that fit. In particular, comparing the best fitting parameters between infants and adults could provide insights into the origin of the developmental differences in behavior (see 'Joint Scaling' under Methods for technical details). When investigating which parameters were most influential in achieving better fit, we found that σ_ϵ played a key role in improving fit. A high σ_ϵ improved fit to the infant data, but a low σ_ϵ improved fit to the adult data (Figure 11). Conceptually, σ_ϵ represents an agent's beliefs about how much noise there is in their own perceptual input, suggesting that infants' learning behavior in our task was adjusted to higher perceptual noise.

4.6 Discussion

Using within-subjects measurements of looking time, we examined the sensitivity of looking to familiar vs. novel stimuli as a function of prior exposure. We did not find evidence for non-linearities predicted by prior models (Fig. 1A & 1B), and instead found that habituation and dishabituation followed monotonic trends as predicted by RANCH (Fig. 1C).

We next test whether RANCH's can make predictions in a novel dataset.

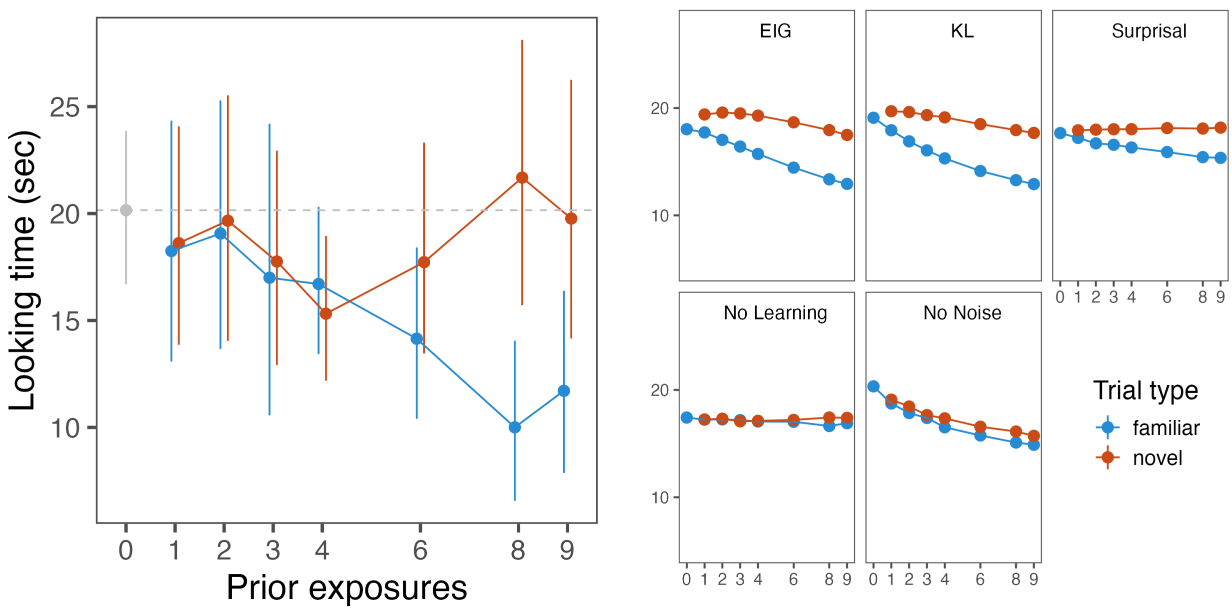


Figure 4: Experiment 1, infant results

5 RANCH predicts stimulus similarity-based dishabituation in infants and adults

Beyond predicting non-linearities in attentional preferences during encoding (Fig. 1B), Hunter and Ames [25] also predicted that after complete encoding of a stimulus, dishabituation magnitude should be linked to the similarity between two stimuli: the more dissimilar the novel stimulus is to the habituation stimulus, the more participants should dishabituate. We designed an experiment testing this prediction by systematically varying the similarity between the familiar and novel stimuli in a block. RANCH provides an explicit notion of similarity through the use of ResNet50-derived image embeddings. We can therefore test whether RANCH also predicts such similarity-based differences in dishabituation magnitude. We further test RANCH by assessing whether the best-fitting parameters in our previous experiment can be used to predict infants' and adults' looking times in a new experiment.

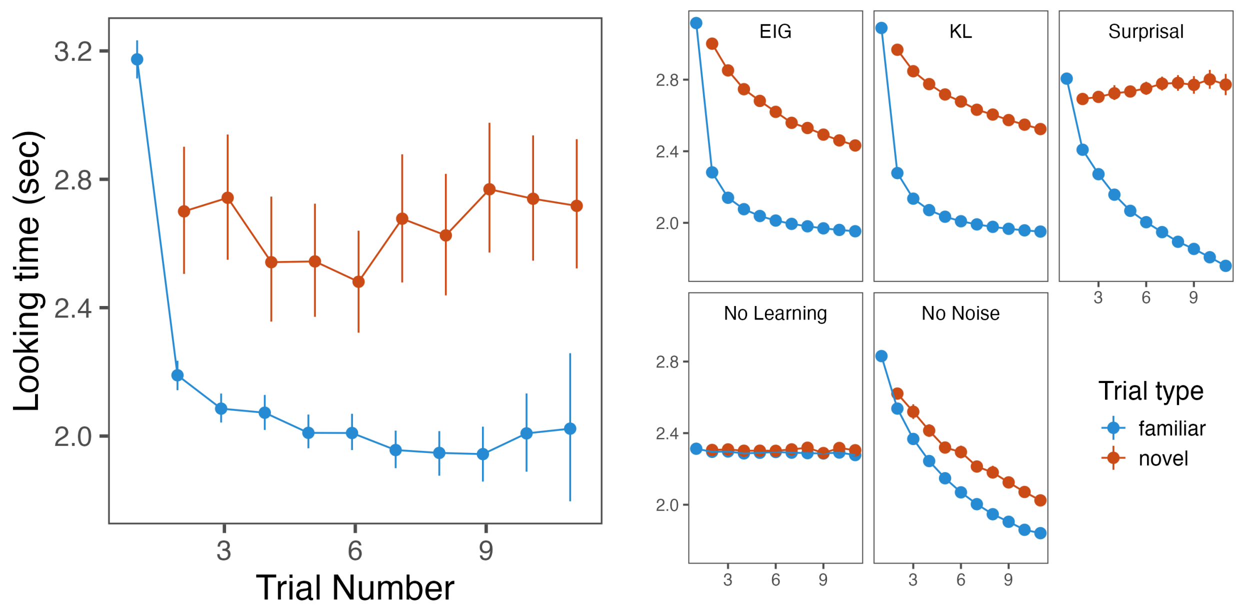


Figure 5: Experiment 1, adult results

5.1 Behavioral Experiment

5.1.1 Infant experiment

To study whether infants' dishabituation was modulated by similarity between the habituation and novel stimulus, we ran an experiment via Zoom (n=57, age range=7-10 months, M_{age}=8.99months), and an exact replication via Children Helping Science, an asynchronous online platform for online developmental data collection [Scott and Schulz [42]; n =66, age range=6-12 months, M_{age}=9.69months). We presented infants with a series of blocks, each again consisting of a familiarization and a test trial (Figure 3, Experiment 2, Infants). Instead of modifying prior exposure as in the previous experiment, here infants were always familiarized 8 or 9 times with an animal (corresponding to the amounts of exposure that caused robust dishabituation in the previous experiment). After familiarization, infants saw a test trial, which could relate to the familiar animal in one of five ways: they either saw the familiar animal again (familiar), the same animal facing in the opposite direction (pose), a different animal facing the same way (identity), two instances of the same animal side-by-side (number), or an inanimate vegetable or fruit (animacy).

In a separate pilot experiment, we also tested baseline interest in these stimuli without prior familiarization, to ensure that condition differences in the main experiments are due to violations, rather than intrinsic to the stimuli (n=35, age range=7-10 months, M_{age}=9.21months). This experiment suggested no baseline differences in looking time between categories (Figure 13).

5.1.2 Adult experiment

We additionally ran an adult experiment that was procedurally similar to the previous adult experiment but whose stimuli were similar to the current infant experiment. Participants (N = 468) saw a sequence of animation, with repeating stimulus and one novel stimulus that appears at the end of the sequence. Since adults have longer attention span, we could present adults with all permutations of the familiar stimulus and novel stimulus, including all of the different levels of similarity in the infant experiment. The familiar stimuli were either animals or vegetables, facing either left or right, presented either singly or in a pair.

5.1.3 Behavioral and model results

Our experiments successfully elicited differential dishabituation in both infants and adults. Infants showed dishabituation to all novel test trials; identity: $\beta = 0.33$; $SE = 0.16$; $t = 2.08$; $p = 0.04$, number: $\beta = 0.32$; $SE = 0.17$; $t = 1.92$; $p = 0.057$, animacy: $\beta = 0.33$; $SE = 0.16$; $t = 2.02$; $p = 0.046$), except the pose violation, which elicited no significant dishabituation (pose: $\beta = 0.13$; $SE = 0.16$; $t = 0.83$; $p = 0.406$).

Our exact replication via Children Helping Science showed similar results: We again found that animacy and number resulted in the strongest dishabituation in infants (animacy: $\beta = 0.34$; $SE = 0.14$; $t =$; $p = 0.015$, number: $\beta = 0.35$; $SE = 0.13$; $t = 2.68$; $p = 0.008$), and pose violations did not elicit dishabituation ($\beta = 0.02$; $SE = 0.13$; $t = 0.14$; $p = 0.888$). However, unlike the Zoom study, the identity violation did not elicit significant dishabituation in our Lookit study ($\beta = 0.07$; $SE = 0.14$; $t = 0.46$; $p = 0.646$). [2]

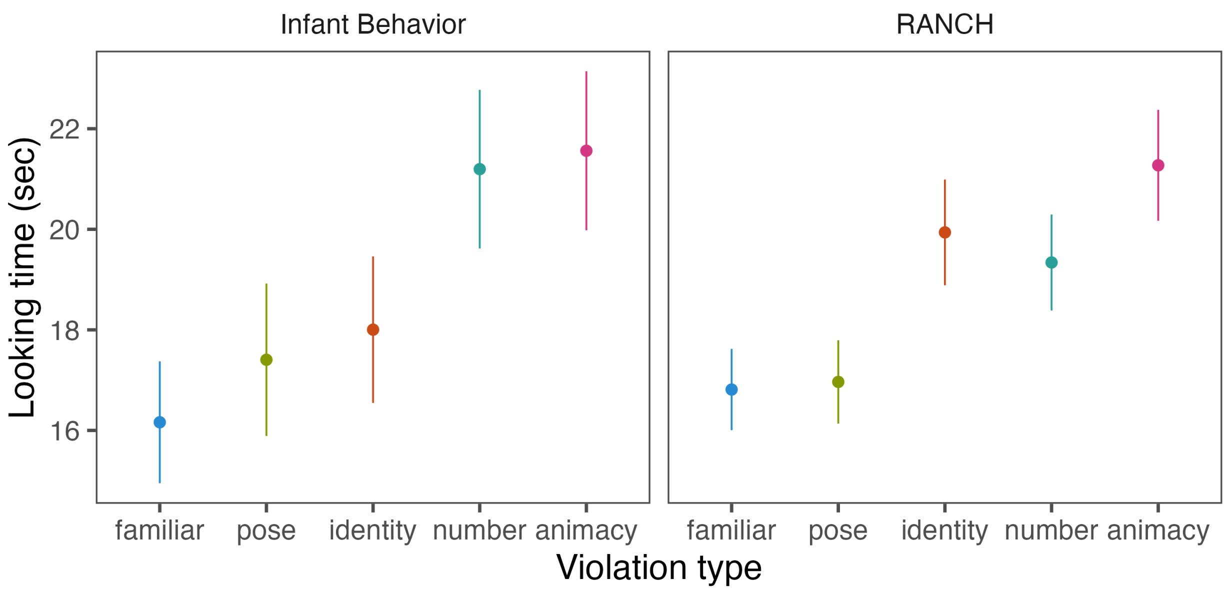


Figure 6: Experiment 2, infant results, combined across Zoom and Children Helping Science

We then combined the two datasets (total n=123) to gain additional power. In an exploratory analysis, we ran the same linear model as on the individual datasets, since we found no additional variance explained by a random effect of dataset. We found again found significant dishabituation to animacy and number violations (number: $\beta = 0.34$; $SE = 0.11$; $t = 3.19$; $p = 0.002$, number: $\beta = 0.35$; $SE = 0.1$; $t = 3.37$; $p = 0.001$), marginally significant dishabituation to identity violations ($\beta = 0.19$; $SE = 0.11$; $t = 1.77$; $p = 0.078$) and no dishabituation to the pose violation ($\beta = 0.07$; $SE = 0.1$; $t = 0.65$; $p = 0.514$). Together, we found evidence that the magnitude of dishabituation was related to the similarity of the novel trial, as infants' looking time followed our expected ordering: animacy > number > identity > pose > background (Figure 6).

Next, we tested whether infants' dishabituation was better described by a "same or different" account, in which dishabituation responds similarly to any novel stimulus, or a "graded" account, in which the type of violation influences the magnitude of dishabituation. To do so, we conducted a model comparison in which violation type was either parsed into two levels (familiar or novel) or five levels (familiar, animacy, number, identity or pose). The model comparison significantly favored the more granular model ($\chi(4) = 8.93$; $p =$

[2]Note that we pre-registered the identity violation as a data quality check, since we had replicated that effect in both infant experiments. However, surprisingly, the identity condition was the only condition that differed from the Zoom experiment, so here we deviate from pre-registration and report these Lookit findings regardless given the high value of having a second sample of infant data.

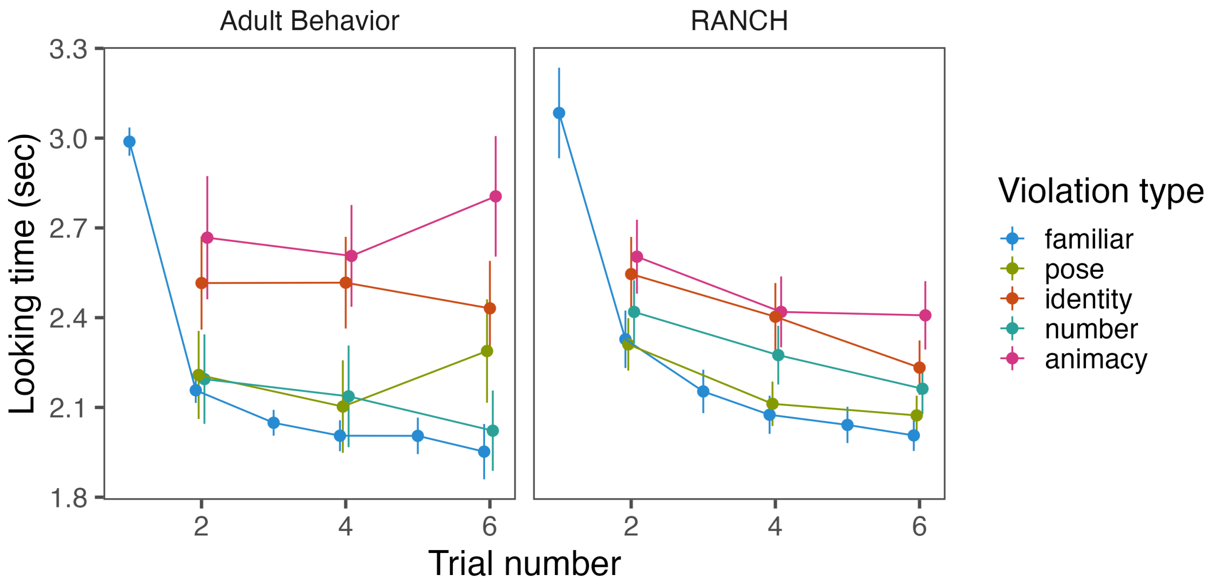


Figure 7: Experiment 2, adult results

0.063), suggesting that infants were sensitive to the type of violation rather than just to whether the test stimulus was familiar or novel.

In adults, as in previous experiments, we found evidence for habituation (Figure 7; β = -0.02, SE = 0, $p <$ .001). More importantly, we also found evidence for graded dishabituation: the dishabituation magnitude to animacy violations was larger than to number violations (β = 0.17, SE = 0.04, $p <$ .001) and to pose violations (β = 0.18, SE = 0.04, $p <$ .001). The same pattern was found for identity violations (β = 0.18, SE = 0.04, $p <$ .001). However, animacy violations were not different from identity violations, nor were number violations different from the pose violations (all p > 0.1). Qualitatively, we found that the ordering for dishabituation magnitude was animacy > identity > pose > and number (Figure 7).

We next tested whether RANCH made similar predictions. We used the modeling procedures for the infant and adult experiments described above.

Using the previously best-fitting parameters in Experiment 1, we found that RANCH generalized to Experiment 2 successfully (Infants: $RMSE$ = 1.2559228; R^2: 0.6553665; Adults: 158.8949134; R^2: 0.7168919). Moreover, RANCH also showed a qualitative ordering of the dishabituation magnitude: animacy > identity > number > pose. This ordering was in line with the CNN-derived embedding distances between the different stimulus categories (Figure 12), and similar to those seen in infants and adults.

Taken together, these data suggest that RANCH is not only capable of capturing habituation and dishabituation, but also can generalize to a new task and make predictions about dishabituation magnitude in relation to stimulus similarity. Some stimulus similarity differed between infants, adults, and RANCH - we discuss these in more depth below.

6 General Discussion

Even though habituation has been a key method for gaining insight into the infant mind, the underpinnings of this behavior are not well understood. Here, we investigated how well habituation and dishabituation can be understood as rational exploration via noisy perceptual samples. To do so, we introduced the RANCH model, which takes noisy samples from stimuli and makes decisions about whether or not to continue sampling based on the expected information gain of a new sample. We used RANCH to make predictions about infants' and adults' looking times across a range of exposures, and the model showed good fit across a range of parameters. In a second experiment, we found that the same parameters in RANCH could be used to make predictions about looking time in a new task where we varied the similarity between habiutation and dishabituation stimuli. Taken together, these results suggest that RANCH provides a quantitative account of looking time as resulting from optimal exploration of noisy perceptual samples.

6.1 Unique strengths of RANCH framework

A key strength of RANCH is the cognitive interpretability of its parameters. Previous Bayesian models of infant attention have leveraged parameter interpretability to draw conclusions about the computational mechanisms underlying individual differences in attention [39]. Here, we use parameter fits of RANCH to study population differences: By applying a "joint scaling" procedure in which we scale model samples to infant and adult simultaneously, we could identify which parameters had to be different between infant and adults to account for their different patterns of looking time. Using this procedure, a key developmental difference was in σ_ϵ, the learner's prior on their perceptual noise. Given that σ_ϵ was preferred to be higher in infants than in adults, this suggests that infants' learning in our experiments may be tuned for noisier perceptual processing.

Another strength is RANCH's ability to model looking on a more fine-grained timescale. Prior research has largely resorted to modeling on the trial-level, and asking about correlations between behavior on a given trial (e.g. looking time) and certain information-theoretic proxies, such as KL or surprisal [28, 38]. While a rational analysis relies on computing expected information gain, this is not possible at the resolution of these prior models, because they carve up time in terms of trials, rather than samples. It is only in moment-by-moment modeling frameworks like RANCH that we can implement the formal rational analysis [34, 1], in which learners ask "How much information do I expect to gain from taking another sample from the stimulus?". The ability to conduct the rational analysis enhances our understanding of habituation as optimally adapted to environmental constraints [see e.g. 18, for a different rational analysis of habituation]. Still, replacing EIG with information-theoretic proxies in our setting can help us examine whether participants' behavior seems to follow an EIG-based, or a proxy-based, linking hypothesis. In our analyses we found that KL divergence performed comparably to EIG, and the surprisal-based model was worse at fitting both infant and adult data in Experiment 1.

6.2 Novel within-subjects designs

To test our model, we developed novel paradigms for within-infant measurements of habituation and dishabituation. These novel designs allowed us to go beyond testing simple qualitative predictions (i.e. longer versus shorter looking duration). Instead, we validated RANCH's continuous predictions about the relationship between prior exposure and looking to novel vs. familiar stimuli (Experiment 1) and between perceptual distance and looking (Experiment 2), within subjects. Scaling this approach through remote infant behavior testing and automated gaze coding via iCatcher+ [15] has allowed us to collect within-subject data from 261 infants across experiments.

6.3 Graded dishabituation in humans and models

Our findings in Experiment 2 warrant further discussion. In particular, the number violation resulted in interesting patterns across age groups and RANCH: The number of samples RANCH preferred to take from number violations was less than identity violations but more than pose violations, similar to adults. Infants however looked at number violations more than identity violations. This discrepancy between the model and adults on the one hand, and infants on the other hand could be explained by a variety of factors.

RANCH is designed for representing single objects, not multiple objects, therefore any dishabituation to number violations results primarily from simple, low-level perceptual distances, rather than dishabituation to number per-se. To the extent that the dishabituation observed in the data exceeds RANCH's, we can say that this points to other, conceptual sources of surprise that cannot be explained purely by low-level visual differences. This is consistent with infant literature showing that infants are sensitive to changes in number specifically, rather than visual changes generally [45, 17]. Adults on the other hand did not show this exaggerated effect of number violations, possibly because adults' processing was shallower in our task, which is supported by their shorter reaction times.

Infants' stronger dishabituation to number violations than predicted by RANCH was replicated in our Children Helping Science experiment. However, unlike in the original Zoom experiment, the Children Helping Science experiment did not elicit significant dishabituation to identity violations. This is in contrast to our findings in both Experiment 1 and Experiment 2, where identity violations after long familiarizations caused longer looking times than looking to the background. There may be two possible reasons for this discrepancy: First, while in the Zoom study experimenters terminated trials when infants looked away for three seconds, the experiment used fixed trial durations due to the asynchronous set-up. While in both experiments looking times are still calculated using a standard criterion off-look [24], the infant-contingency of the Zoom experiment likely increases the data quality and results in higher sensitivity to smaller effects. Similarly, an experimenter being present via Zoom allows them to ensure a quiet testing environment and adjust the flow of the experiment to the participant - previous comparisons between Zoom and asynchronous studies similarly suggest that effects via Zoom tend to be larger.

6.4 No evidence for familiarity preferences

A key feature of our experimental data, as well as our model, is that at no point, even for the shortest familiarization durations, we find longer looking to familiar than to novel stimuli. This finding contrasts with predictions made by a prominent model of infant attention (Figure 1A)), which posit that at intermediate levels of encoding, familiar stimuli should be more interesting than novel stimuli. The intuition is that some initial familiarity with a stimulus may allow a viewer to "break into" the information there is to gain from the stimulus - a completely novel stimulus may be too complex or foreign for extended attention to be worth it. In the learning setting of simple perceptual learning, the rational analysis and participants do not seem to favor this strategy. This suggests that limited exposure does not generically lead to familiarity preferences in any experimental paradigm.

Of course, the absence of familiarity preferences in our results does not rule out their existence in general. First, familiarity preferences may be more subtle than novelty preferences, so that the statistical power that is needed to find familiarity preferences is higher than that achieved in the current study. A current large-scale study by the ManyBabies consortium which aims to test the predictions made by Hunter and Ames [25] may give insight into this possibility [29]. In addition, the learning problem that participants are solving likely plays a critical role in whether they will exhibit familiarity preferences. This context-dependence is reflected in meta-analyses investigating familiarity preferences in different paradigms. For example, when tested on word segmentation in their native language, infants show persistent preferences for familiar stimuli throughout the first year [5]. In contrast, when tested on statistical learning of novel words, infants show consistent preferences for novel stimuli, from 4 months to 11 months of age [6].

6.5 Formal theories of learning and attention

These seemingly contradictory results highlight the need for formal theories of how the learning problem influences attention. Dubey and Griffiths [14] give an example of such a formal account by showing that when past and present events are correlated, rational agents, under some assumptions, develop a tendency to attend to familiar stimuli to prepare for the most likely future events, while in uncorrelated environments, novelty preferences are optimal. On the other hand, ideal learners attempting to maximize their expected information gain consistently seek novelty when trying to learn a single concept. Formal models of the learning problem being solved in experiments may therefore give more principled and precise predictions beyond identifying factors like "prior exposure" as determinants of attentional preferences.

Importantly, RANCH's predictions outlined in this paper apply to the specific case: the learner makes noisy observations of 1) stimuli from a specific embedding space, 2) updates its beliefs about the location of a single concept and 3) makes sampling decisions based on EIG. The modular structure of RANCH allows for each of these components (embedding space, learning model and linking hypothesis) to be modified, likely leading to different predictions. For example, the embedding space could be chosen to more closely resemble a psychologically validated perceptual space [e.g. 31], though in this case, we found that such alternatives did not affect embedding structure qualitatively (Figure 12)). Similarly, the learning problem could be modified to account for more complex, or more general, representation learning [26, 43]. For example, in our task participants may not just be learning a single concept, but rather also decide whether a stimulus should prompt the creation of a new concept or not.

6.6 Conclusion

From birth, infants actively explore their environment through selective attention. Looking time, a widely used measure in developmental psychology, has long been used to make inferences about infants' conceptual and perceptual capacity. Here, we measure nuances of key looking time phenomena, such as habituation and dishabituation, by varying familiarity with a stimulus prior to dishabituation (Experiment 1), as well as the similarity between the familiarization and test stimulus (Experiment 2). We find that these behavioral nuances are well captured by RANCH, an image-computable, rational model that learns from noisy perceptual samples and decides how many samples to take based on their expected information gain. Beyond modeling infant looking during perceptual learning in particular, our work provides a general framework in which to instantiate and test hypotheses about computations underlying infant looking.

7 Methods and Materials

7.1 Model Specifications

7.1.1 Perceptual representations

We derived principled low dimensional representations of our stimuli from ResNet50 [22]. We ran a Principal Component Analysis and used the first three principal components. This decision was made due to the computational demands of our model. Each added dimension would increase the total run time exponentially. The first three components captured 57.9% of the variance.

Given that we were using discrete grid approximation in our inference, the model was sensitive to the absolute values of the embeddings. When these embeddings values were close to or exceeded the range of the approximation grids, this would cause bias in our simulation result. We therefore scaled our embeddings down to fall squarely into our grid. This downscaling had to be more extreme for the stimulus type experiment, where embedding distances were larger, in particular for animacy violations. We ran simulations to confirm that embeddings that were too large would result in biased model behaviors (Figure 10).

7.1.2 Linking Hypotheses

We compared two additional information theoretic measures with Expected Information Gain (EIG): Kullback–Leibler divergence and Surprisal. Each of these measures has been invoked in the literature as a plausible account for explaining looking behaviors across developmental stages [KL divergence: 37, surprisal: 28].

Our key linking hypothesis, EIG, is a forward-looking measure. It is computed as the product of the posterior predictive probability of the next perceptual sample ($p(z_{t+1}|\theta_t)$) and the information about the concept *theta* gained conditioned on that next hypothetical sample, using a grid of possible subsequent samples (S): $EIG(z_{t+1}) = \sum_{z_{t+1} \in S} p(z_{t+1}|\theta_t) * D_{KL}(\theta_{t+1}||p(\theta_t))$

Both KL-divergence and surprisal are backward-looking measures. The KL divergence at any given point is calculated by the posterior distribution after seeing the current perceptual sample and before seeing the current perceptual sample: $D_{KL}(\theta_t || p(\theta_{t-1})) = \int p(\theta_t) log \frac{p(\theta_t)}{p(\theta_{t-1})} d(\theta)$

Surprisal is calculated as the negative log likelihood of the current perceptual sample under the prior distribution: $surprisal(z_t) = -log(p(z|\theta_{t-1}))$

We ran individual simulation and parameter search for each of the three linking hypotheses to ensure fair comparison.

7.1.3 Scaling Procedure for Developmental Interpretation

To find the best fitting parameters for each age group we employ a strategy we call "joint scaling". To evaluate model fit, we first map model samples onto looking times by finding the best linear scaling of model samples in a regression of the form `LT ~ model_samples`. The parameter set that minimizes the RMSE between the scaled model samples and looking times is what we consider the best parameter set. In the case of "joint scaling", we regress both the adult and the infant simulations against their looking times simultaneously, to find how varying the infant and adult parameters affect the fit of the scaled model samples to the adult and infant data. This joint scaling procedure puts pressure on the parameters, rather than the scaling, to account for the fact that adults' looking times are a lot shorter than infants' looking times.

The interpretable parameters of RANCH are the priors on μ and σ and the priors on the noise. The priors on μ and σ were parameterized by a normal inverse-gamma prior with μ_{prior}, ν_{prior}, α_{prior} and β_{prior}, the conjugate prior to a normal distribution with unknown mean and variance. While the mean was fixed to be 0 (μ_{prior}), variation in the other parameters express distinct hypotheses about the precision of the location of the concept (ν_{prior}), as well as the variance of the concept (α_{prior} and β_{prior}). The learner's prior on noise are parameterized by gaussian distribution with a mean (μ_{noise}) and standard deviation (σ_{noise}). In our parameter search, we only searched over the σ_{noise}.

7.2 Experimental Procedure

Below we describe in detail the experimental procedure. We use Experiment 1 to denote the experiment that manipulates exposure duration, and Experiment 2 to denote the experiment that manipulates stimuli similarity.

7.2.1 Infant experiment

7.2.1.1 Participants For Experiment 1, we tested a combined sample of 103 7-10 month old infants, with 31 in Exp A, 35 in Exp B and 37 in Exp C (103 (M_{age} = 9.38 months, 47 female). 10 participants were excluded completely due to fussiness. An additional 58 individual test events were excluded (out of 552.

For Experiment 2, we tested a combined sample of 123 7-10 month old infants, with 57 in the Zoom experiment (7 complete exclusions, 42 trial exclusions), 66 in the Zoom experiment (3 complete exclusions, 26 trial exclusions) and 35 in the control experiment (2 complete exclusions, 10 trial exclusions).

Complete or trial exclusions occurred because 1) infants fussed out of the experiment at an earlier stage of the experiment, 2) infants looked at the stimuli for less than a total 2 seconds, 3) there were momentary external distractions in the home of the infant or 4) the gaze classifier performed so poorly that manual corrections were not viable.

7.2.1.2 Stimuli In Experiment 1, we presented infants with a series of animated animals, created using "Quirky Animals" assets from Unity (Figure 3), link to assets). The animals were walking, crawling or swimming, depending on the species. In a previous set studies, not included here (but see OSF repository), we showed infants a different stimulus set and failed to elicit replicable habituation, novelty or familiarity preferences. As a result, we modified the stimuli to be more appealing, and separated familiarization trials with curtains opening and closing to accentuate familiarization trials as separated occurrences of the stimulus.

In Experiment 2, (Figure 2A, we introduced animacy violations using the “3D Prop Vegetables and Fruits” link to assets) assets from Unity and animated bouncing and rotating trajectories for them. To create number violations, we duplicated the videos of single animal Unity assets and pasted them side-by-side.

7.2.1.3 Procedure Data collection was performed primarily synchronously on Zoom, except for the Lookit replication of Experiment 2. In all cases, infants were recruited from Lookit [42] and Facebook. Parents were instructed to find a quiet room with minimal distractions, place their child in a high chair (preferred) or on their lap, and to remain neutral throughout the experiment. Infants were placed as close as possible to the screen without allowing them to interact with the keyboard.

Both main experiments followed a block structure, where each block was divided into two sections: 1) a familiarization period and 2) a test event. Each block was preceded by an “attention getter”, a salient rotating star. During the familiarization period, the infant was familiarized to a particular animal, the background, in a series of familiarization trials. Each familiarization trial was a 5 second sequence: curtains open for 1 second, the animated animal moves in place for 3 seconds, and then the curtains close for 1 second. In the exposure duration experiments, the number of familiarization trials (the “exposure duration”) varied between blocks. In the stimulus type experiment, the number of familiarizations was always 8 or 9.

During the test event, the infant saw either the same background animal again, or a novel stimulus, the deviant. The onset of the test event was not marked by any visual markers, but a bell sound played as the curtains opened, to maximize the chance of engagement during the test event. On Zoom, the test event used an infant-controlled procedure: the experimenter terminated the trial when the infant looked away for more than three consecutive seconds. On Lookit, test trials would always be played for 40 seconds. In both cases, looking time was then defined as the total time that the infant spent looking at the screen from the onset of the stimulus until the first two consecutive seconds of the infant looking away from the screen. If the infant did not look away after 60 seconds of being presented with the test event, the next block automatically began and infants’ looking time for that test event was recorded as 60 seconds.

In Experiment 1, the deviant stimulus was always another animal. In Experiment 2, one of four properties of the background animal could be violated: pose, identity, number or animacy.

In Experiment 1, each infant saw six blocks: Three different exposure durations (0, 4 and 8 in Exp. A, 1, 3 and 9 in Exp. B and 2, 4 and 6 in Exp. C) appeared twice each, once for each test event type (background or deviant). The longer exposure durations (8 or 9) were chosen based on our previous pilot studies with a different stimulus set (OSF repository), and the shorter durations were chosen to provide limited learning experience with the background. The order of blocks was counterbalanced between infants, and pairs of animals (background and deviant) were counterbalanced to be associated with each block type. Which animal was shown as the background and which as the deviant (if it was a deviant test trial) was randomized in each block.

In Experiment 2, each infant saw four blocks within a session - the background trial was always one of the four test trials, and the remaining three test trials were picked randomly from the remaining four violation types (pose, identity, number, animacy). The control experiment showed six standalone trials: Infants saw two single animals, two pairs of animals and two single vegetables, to measure baseline interest in these stimulus categories.

7.2.1.4 Looking time coding To code the infants’ looking time we used iCatcher+, a validated tool developed for robust and automatic annotation of infants’ gaze direction from video [15]. To obtain trial-wise looking time, we merged iCatcher+ annotations with trial timing information, thereby fully replacing manual coding of looking time. The correctness of iCatcher+ coding was manually supervised to check for failed face detections or distractions.

7.2.1.5 Statistical analysis

We ran the pre-registered statistical analyses for both experiments. All statistical analyses were ran in R using the package `lme4` [4].

For Experiment 1, we ran a `lme4` model with the following specification: `log(looking time) ~ poly(exposure duration, 2) * test type [identity or background] + log(block number) + (1|subject)`. The `exposure duration` term refers to the amount of prior exposure received by the infant (0 to 9 prior exposures), and the second polynomial tested for non-linearity in the effect on looking time, as predicted by some theories of infant attention. This non-linearity could occur primarily for looking to the background (Figure 1A), or both background and novel stimuli (Figure 1B), hence the interaction with `test type`. We control for `block number` to account for general decreasing interest in our stimuli as the experiment goes on.

For Experiment 2, we ran a linear mixed effects regression, again predicting looking time, using violation type as the main predictor: `log(looking time) ~ test type [animacy, pose, identity, number, background] + log(block number) + (1|subject)`. Here, we set `background` as the reference level for `test type`, and asked whether the other test types were significantly different from that background reference level. We again controlled for block number.

7.2.2 Adult experiment

7.2.2.1 Participants

Both experiments were run on Prolific (Experiment 1: N = XXX; Experiment: N = XXX). Two experiments have the same pre-registered exclusion criteria. We excluded participants if (1) the standard deviation of their reaction time across all trials was less than 0.15 (indicating key-smashing, Experiment 1: N = XXX; Experiment 2: N = XXX), (2) spent more than three absolute deviations above the median of the task completion time as reported by Prolific (Experiment 1: N = XXX; Experiment 2: N = XXX), and (3) provided the wrong response to more than 20% of the memory task (Experiment 1: N = XXX; Experiment 2: N = XXX). A total of XXX and XXX participants met at least one of the three criteria and were excluded from the final analysis for Experiment 1 and Experiment 2, respectively. We also excluded a trial if the trial was three absolute deviations away from the median in the log-transformed space across reaction time from the remaining participants.

The final sample included XXXX participants (Experiment 1: Mean Age, SD Age; Experiment 2: Mean Age, SD Age).

7.2.2.2 Stimuli

The stimuli used in adult experiment are identical to the ones used in infant experiment. In Experiment 1, the stimuli are selected from the animal unity set. In Experiment 2, both the animal and vegetable sets were used.

7.2.2.3 Procedure

Procedurally, Experiment 1 and Experiment 2 are similar. Both are self-paced visual presentation studies. Participants were instructed that they would be looking at a sequence of animated stimuli at their own pace. At each trial, then can press a key on the keyboard to move on to the next trial after a minimum viewing time of 500 ms. Each study consists of multiple blocks. Each block consists of multiple trials. Between blocks, participants answered a simple memory question ("Have you seen this animation before?"). This memory question is used as an attention check.

Each block contains multiple trials of one stimulus being repeatedly presented (the familiar trials) and one trial that shows a different stimulus (the novel trial). The novel trial always appears as the last trial. In Experiment 1, the total number of trials range from two to eleven. The novel trial was randomly drawn from the stimuli set that researchers haven't seen before. In Experiment 2, the block consists of either two, four, or six trials. There are a total of eight types of stimuli in the familiar trials. They are all combinations of the three features that each include two levels: animacy (e.g. animate or inanimate), number (singleton or pair), and pose (facing left or facing right). The novel trial can differ from the familiar trials in one of the four dimensions: pose, number, identity, and animacy. Both experiments also have blocks that only include one stimulus (the familiar block).

The order of the block is semi-randomized in both experiments. Experiment 1 has 15 blocks in total. We grouped the total number of trials in each block into five pairs of two: [2, 3], [4, 5], [6, 7], [8, 9], [10, 11], and randomly sampled 1 block length from each pair. In other words, each participant would receive 5 different lengths of repeating background trials, stratified by 5 levels. We controlled the distribution of the block lengths by making sure the first half and the second half of the experiment each has the same number of blocks with different block lengths. Experiment 2 had 24 blocks in total. We grouped the twenty-four blocks into four groups. Each group consisted of two familiar blocks and one block from each of the four novelty types. The order of blocks within each group was randomized.

7.2.2.4 Statistical Analysis We ran the pre-registered statistical analyses for both experiments. All statistical analyses were ran in R using the package `lme4` [4].

For experiment 1, we ran a linear mixed effect regression predicting the log transformed of looking time with the following specification: `log(looking time) ~ trial_number + is_first_trial + trial_number * trial_type + log(block_number) + (trial_number * trial_type | subject) + (is_first_trial + trial_number | subject)`. This model allows us to investigate whether the number of familiar trials has an impact on the looking time (i.e habituation), and whether the trial type (familiar or novel) mediates the influences. This model failed to converge. Following the pre registered procedure, we pruned the model to include only by-subject random intercept. The pruned model suggests that there is evidence for habituation and dishabituation (STATS).

For experiment 2, we ran a similar model and incorporated the effect of different novelty type. The particular model specification is as follows: `log(total_rt)` $\sim$ `trial_number + is_first_trial + (trial_number + is_first_trial) * stimulus_number + (trial_number + is_first_trial) * stimulus_pose +(trial_number + is_first_trial) * stimulus_animacy + (trial_number + is_first_trial) * novelty_type + log(block_number)`. The `novelty_type` has five levels, including the background trial and four types of violation.This model can help us test two hypotheses: (1) whether our experimental paradigm captured habituation and dishabituation and (2) whether the magnitude of dishabituation was influenced by the similarity between the novel trials and the familiar trials.

8 Supplementary Information

8.1 Deriving predictions for Goldilocks model

Below is an illustration of how predictions for a habituation/dishabituation experiment were derived for the Goldilocks model. In all cases, as infants get familiarized, the surprisal of the familiar stimulus goes down, and the surprisal of the novel stimulus has to go up, so that probabilities assigned to events add up to 1. Depending on the initial surprisal of the familiarization stimulus, there are three different trajectories of preferences to novel vs. familiar stimuli as a function of prior exposure: 1) If initial surprisal is below the optimal level to maximize attention, the model predicts an increase, and then a decrease in novelty preferences. 2) If initial surprisal is at the optimal level, then no preference is predicted, assuming a symmetric U-shape. 3) If initial surprisal is higher than optimal surprisal, the model predicts an increasing, and then decreasing familiarity preference.

8.2 Parameter robustness

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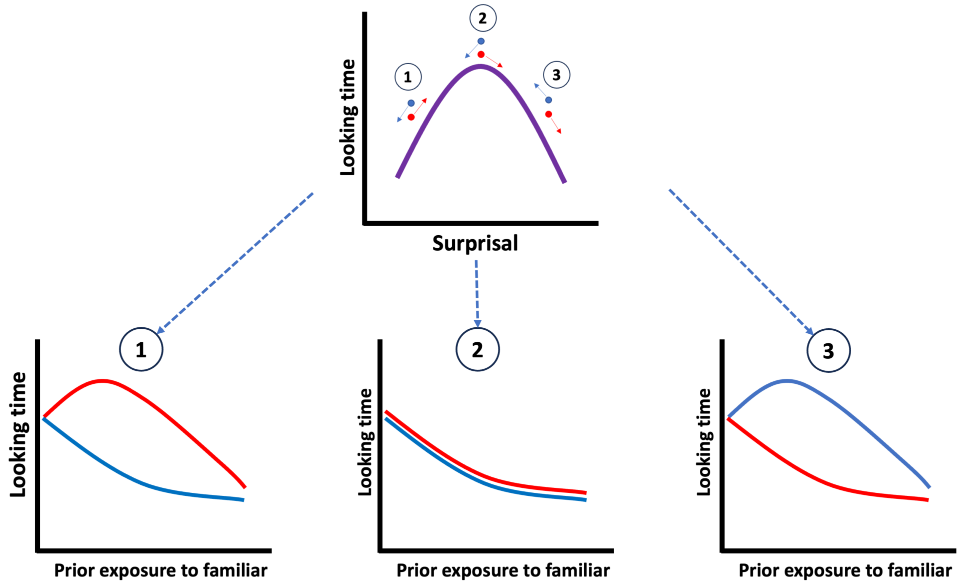

Figure 8: Illustration of how predictions from the Goldilocks model in Fig. 1 were derived.

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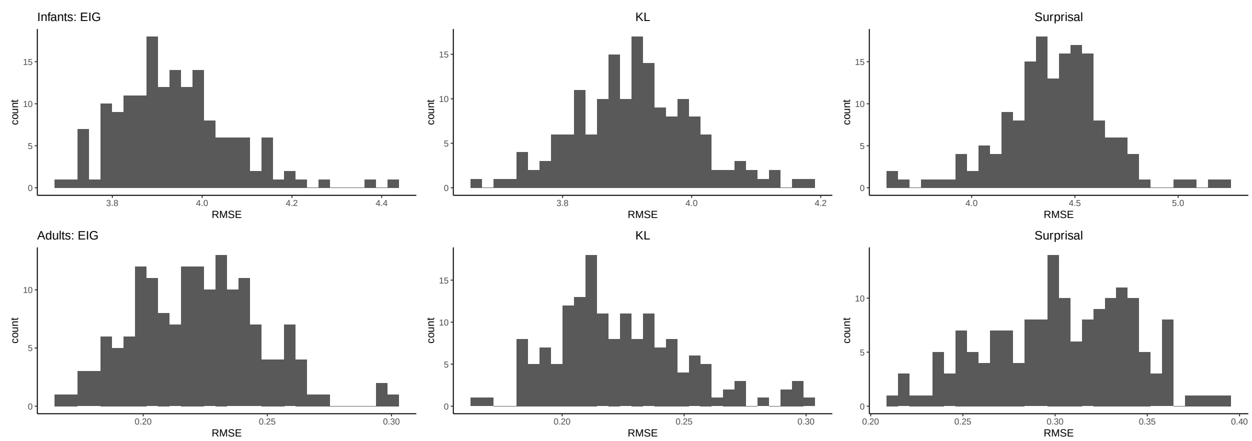

Figure 9: Histograms of RMSE for different linking hypotheses show that RANCH is largely robust to variations in parameters.

8.3 Scaling the embedding space

In order to achieve sensible results, it was important to bring the stimulus embeddings in line with the approximation grid used for inference. Below we show what happens embeddings exceeded or came close to the edges of the our approximation grids. In Experiment 1, unscaled embeddings resulted in a reversal of background vs. deviant model samples, such that the model takes more samples from the background than the deviant. In Experiment 2, unscaled embeddings resulted in an unintuitive ordering which is not in line with the embedding distances (Figure @ref(fig:embedding_distances)).

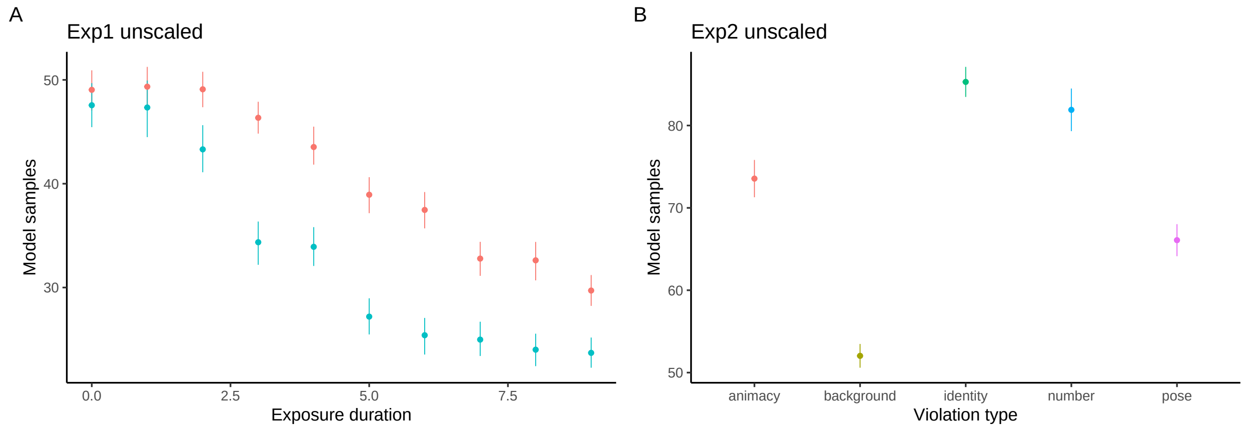

Figure 10: The effect of scaling on model results. A) Experiment 1 with unscaled embeddings results in a reversal of background vs. deviant. B) Experiment 2 with unscaled embeddings results in an unintuitive ordering.

8.4 Developmental comparison of parameter fits

A naive approach to comparison parameter fits between infants and adults would be the following: After joint scaling, pick the set of parameters for adults and infants that together minimize RMSE. If we do this, we obtain the following parameter sets:

infants: ν= 1, α= 1, β= 0.1, σ_ϵ= 1, σ_ϵ= 1, EIG_{env}= 10^{-5}

adults: ν= 3, α= 10, β= 0.1, σ_ϵ= 0.1, σ_ϵ= 0.1, EIG_{env}= 10^{-4}

If we compare these two sets of parameters to each other, and interpret these differences, we may come to the following conclusion: RANCH fits best when infants' σ_ϵ is higher than that of adults, and β_prior is higher than that of adults, suggesting that infants both expect their perception to have more noise, and that infants have more prior uncertainty about the size of the standard deviation.

However, a direct comparison between these parameter sets does not give insight into the relative stability of RANCH's fit to the data if we change any of these parameters: It may be that if a parameter was slightly different, it would not make a big difference for model fit.

In order to investigate this, we want to also check the sensitivity of the model fit to changes in the parameters. To do so, we plot RMSE histograms colored by the value of a certain parameter, from a certain age group. Here we see that the overall modes of the RMSE histograms are dominated by `world EIG`, likely because this parameter is the main threshold that can account for the difference in scale between infant and adult looking times. However, of the other parameters, we see that RMSE only shows sensitivity to σ_ϵ. Other parameters that are different between the best fits do not show this sensitivity. We therefore conclude that the main interpretable parameter difference is in σ_ϵ, the learner's noise prior, being higher in infants than in adults.

8.5 Embedding distances of different violations categories, by embedding model

Below we plot the embedding distances of the different violations. The lines connect a single stimulus, and the y-axis represents the embedding distances to all the stimuli that would constitute a certain violations. For example, for a single animal stimulus, the y-value for 'animacy' corresponds to the distance between the single animal and all the single fruit/vegetable stimuli.

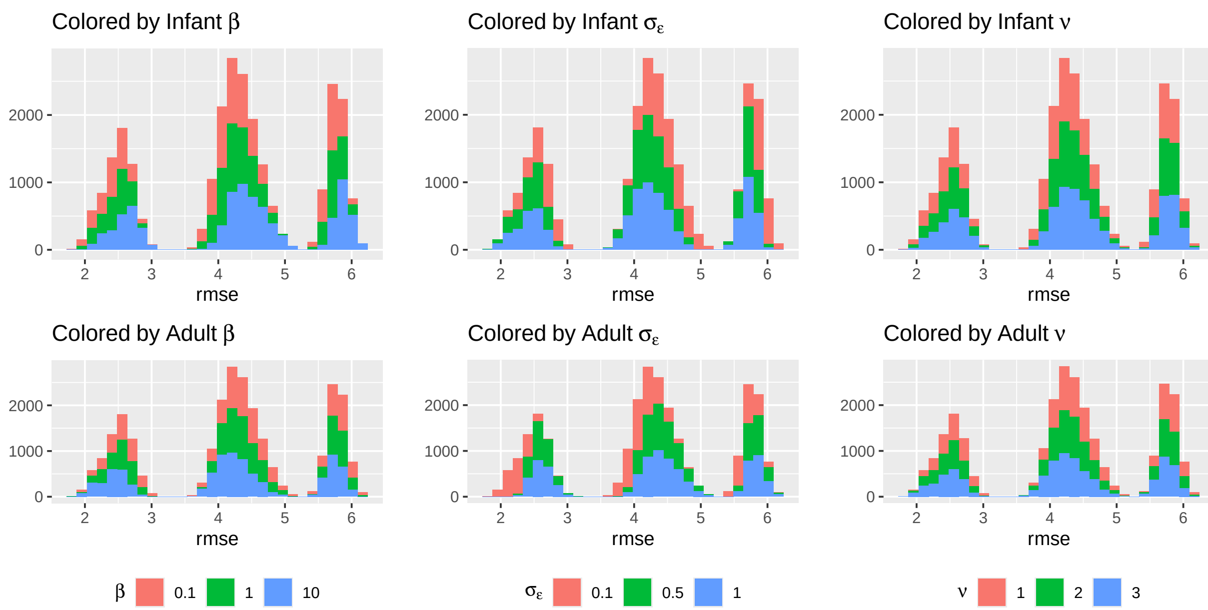

Figure 11: Histograms of model fit colored by priors. Results suggest that when the learner's noise is high, RANCH fits infant data better.

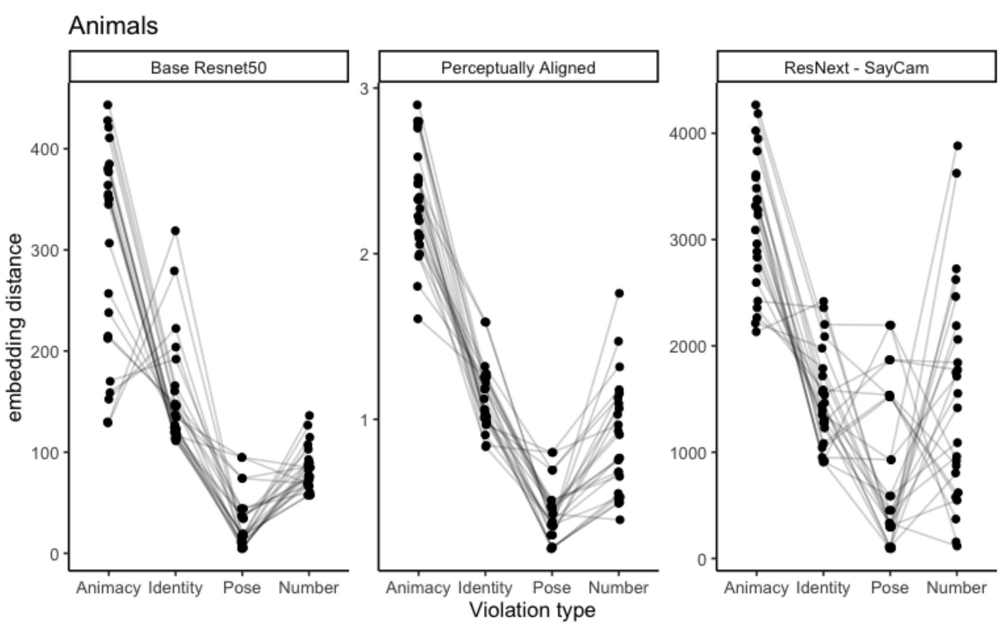

Figure 12: Embedding distances of different violation types for 1) a base ResNet50 model, 2) a perceptually aligned embedding model (Lee, 2022) and 3) a ResNet50 model trained on SAYCAM data (Orhan et al., 2020)

8.6 Control experiment for Experiment 2

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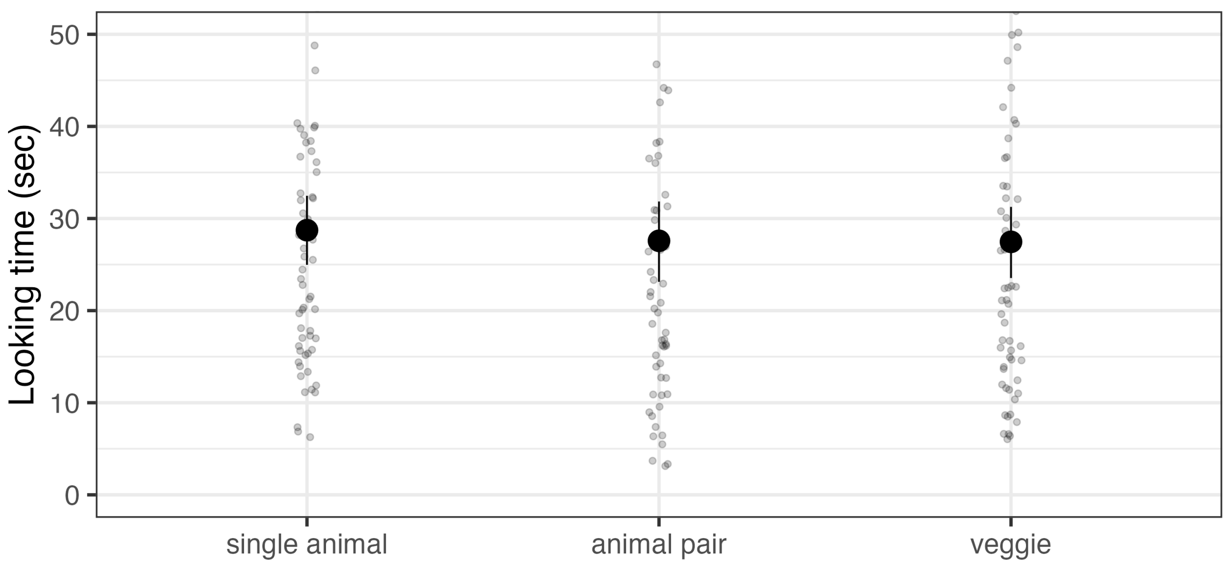

Figure 13: Control experiment for experiment 2, checking for baseline differences in interest.

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